sentiment_analysis_report

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1 Sentiment Analysis Report

1.1 Dataset Description

The dataset used in this project is the Datafiniti Amazon Consumer Reviews of Amazon Products (May 2019), available on Kaggle. It contains thousands of customer reviews of Amazon products across various categories. For this analysis, we focused specifically on the reviews.text column, which includes the actual written feedback from customers.

- File name: Datafiniti_Amazon_Consumer_Reviews_of_Amazon_Products_May19.csv
- Main column used: reviews.text
- Total records before cleaning: Over 34,000
- Text type: Natural language user reviews

1.2 Preprocessing Steps

Several preprocessing steps were performed to prepare the review text for sentiment analysis:

- Missing value removal:
  - Removed rows where reviews.text was NaN.
- Text normalization:
  - Converted text to lowercase using .lower()
  - Stripped leading/trailing whitespace using .strip()
  - Removed punctuation and special characters using regular expressions
- Stop word removal:
  - Using spaCy's token.is_stop attribute, common stop words like "the", "and", "is", etc., were filtered out
- Token filtering:
  - Only tokens contributing meaningful sentiment were retained before analysis
- SpaCy pipeline setup:
  - Used en_core_web_md model
  - Added spacytextblob extension to leverage TextBlob sentiment scoring

1.3 Evaluation of Results

The model was tested on several random product reviews from the dataset. Each review was processed to return a polarity score and corresponding sentiment label:

- Polarity score: ranges from -1 (negative) to +1 (positive)
- Sentiment label: derived based on polarity:
  - Polarity $> 0 \rightarrow$ Positive
  - Polarity $< 0 \rightarrow$ Negative
  - Polarity $= 0 \rightarrow$ Neutral
- **Example Output:**
  - **Review:** I love this tablet. It's super fast, the screen is gorgeous, and it's great for watching videos.
  - **Sentiment:** Positive (Polarity: 0.50)
  - **Review:** Battery life is terrible. Completely useless for travel or long use.
  - **Sentiment:** Negative (Polarity: -0.65)
- The predictions aligned well with human intuition in most cases.

1.4 Model Strengths and Limitations

Strengths

- Easy to implement using spaCy and spacytextblob
- Handles large volumes of text efficiently
- Outputs both a sentiment label and a numeric polarity score
- Pretrained model saves time and training effort

Limitations

- No fine-tuning: The model is general-purpose and not specialized for Amazon reviews
- Subtlety detection: Sarcasm, irony, or mixed sentiment within a review may confuse the polarity score
- Imbalance sensitivity: It may lean toward "Positive" if the review text is vague or lacks strong sentiment
- No context awareness: It doesn't consider metadata like product type or user ratings